

Mockito

- Mockito is a Java framework that creates fake obj called Mock objects for testing
- It allows us to test one class independently without using real dependency.
- In short Mockito is a java framework used to create mock objects for unit testing.
- Suppose we have two services
i.e ① OrderService
② PaymentService

Now if I want to test place orders from PaymentService.

So the Problem & flow is:

Every test

↓
PaymentService

↓
calls bank

↓
takes time

It is time consuming &
not good approach for
testing

- Instead we can do,

orderService



fakePaymentService

↓
output.

This fake object is created using Mockito

Important Terminology

① Real object

ex. : `Payment P = new Payment();`

This is actual class

② Mock Object

ex. : `Payment P = mock(Payment.class);`

This object is fake
It won't execute original code.

③ Stub

→ Means wherever that method is called, return this value

ex. when (payment.pay(500))

• then return (true);

Means,

if pay(500) is called
don't execute original method
simply return true.

④ Verify

→ Means check whether a method was called.

ex. verify(payment).pay(500);

means,

was pay(500) called?

Yes ✓

No X

First mockito program

```
class Demo {
```

```
    public static void main (---) {
```

```
        Payment P = mock(Payment.class);
```

```
        Savt(Payment.pay(500));
```

```
    }
```

O/P: false.

Default values return by Mockito

Ret type	Def values
int	0
long	0
double	0.0
float	0.0f
char	'\0'
String	null
Object	null
List	null
Boolean	false

ex.,

```
class calc {
```

```
    int add(int a, int b) {
```

```
        return a+b;
    }
```

```
public static void main(...) {
```

```
    calc c = mock(calc.class);
```

```
    sout(c.add(5, 4));
}
```

o/p: 0

Beoz mockito never
execute original method.

Calc c = moc(calc.class);
when (c.add(5,4)).
then Return(9);

Now,

cout (c.add(3,4))

O/P = 9